

Epic: Location Dashboard & Viewing

User Story 1: View Location Dashboard

- **As a** Charge Network Operator (CNO),
- **I want** to see a dashboard listing all created locations with their key operational and financial details,
- **so that** I can get a quick, comprehensive overview of the entire charging network infrastructure.
- **Acceptance Criteria:**
 - Given I am logged in, when I navigate to the Locations module, I must see a dashboard table.
 - The table must display the following columns by default: Actions, Organization, Location, Address, Business Category, Reseller, Utility Name, Coordination Node ID, Sales Tax, CNO Fee Percent, CSR Fee Percent, CNO Session Fee, and CSR Session Fee.
 - The data displayed must be the most current information available in the system.

User Story 2: Filter Location Dashboard

- **As a** Charge Station Operator (CSO),
- **I want** to filter the locations list by Organization, Location, Reseller, Utility Name, or Coordination Node ID,
- **so that** I can quickly find specific sites without scrolling through the entire list.
- **Acceptance Criteria:**
 - The dashboard must contain a filter panel.
 - When I enter or select a value in one of the specified filter fields, the list should update in real-time to show only matching locations.
 - I should be able to apply multiple filters simultaneously.

User Story 3: Export Location Data

- **As a** CNO,
- **I want** to export the complete list of location details to an Excel file,
- **so that** I can perform offline analysis and create custom reports.
- **Acceptance Criteria:**
 - There must be an "Export to Excel" button on the Location Dashboard.
 - Clicking the button must trigger a download of a `.xlsx` file.
 - The exported file must contain all location data, respecting any active filters.

User Story 4: View Locations on a Map

- **As a** CSO,
- **I want** to switch to a map view on the dashboard,

- **so that** I can visualize the geographical distribution of my charging sites.
- **Acceptance Criteria:**
 - The dashboard must have a "Map View" toggle or button.
 - Activating map view should display a map with pins representing each location's geographical coordinates.
 - I should be able to interact with the map (zoom, pan) and click on a pin to see key location details.

User Story 5: View Full Location Details

- **As a user,**
 - **I want** to view the full, read-only details of a selected location,
 - **so that** I can review its complete configuration without the risk of making accidental changes.
 - **Acceptance Criteria:**
 - An option to "View" a location must be available in the "Actions" column.
 - The view should display all details, including address, reseller, utility, and network fee information.
 - All fields in this view must be non-editable.
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Epic: Location Creation & Configuration

User Story 6: Initiate Location Creation

- **As a CSO,**
- **I want** to start the "Add Location" workflow and fill in the initial location details,
- **so that** I can begin the process of defining a new physical charging site.
- **Acceptance Criteria:**
 - The "Organization" drop-down must only show CSO-type organizations.
 - The "Location Name" field must be mandatory and accept 3-30 characters.
 - The "Business Category" and "Reseller" drop-downs must be optional.
 - Saving this segment must successfully proceed to the Address segment.

User Story 7: Add Location Address with Map

- **As a CSO,**
- **I want** to add an address for my new location using an interactive map with a search feature,
- **so that** the address is accurate and the geographical coordinates are auto-populated.
- **Acceptance Criteria:**
 - The interface must include a map and an address search bar.
 - As I type an address, the system should suggest matches.

- Upon selecting an address, Latitude, Longitude, City, State, Country, and Zip must be auto-populated and non-editable.
- "Address Line 1" must be mandatory, while "Address Line 2" is optional.

User Story 8: Add Utility Details

- **As a CSO,**
- **I want** to add utility and energy supplier information to the location,
- **so that** energy costs and operational data are correctly associated with the site.
- **Acceptance Criteria:**
 - "Utility Name" must be a mandatory drop-down field fetched from the Utility Module.
 - "Utility Cost" must be a mandatory field that allows decimal values up to three places.
 - "Meter ID" and "Coordination Node ID" must be optional text fields.
 - After saving, the new location must appear at the top of the location listing page.

User Story 9: Edit Existing Location

- **As a CSO,**
- **I want** to edit any field of an existing location across all three segments (Details, Address, Utility),
- **so that** I can keep my site information accurate and up-to-date.
- **Acceptance Criteria:**
 - An "Edit" option must be available in the "Actions" column for each location.
 - The edit workflow must present the same fields and segments as the creation workflow.
 - All validations from the creation flow (e.g., mandatory fields, character limits) must apply during the edit process.

Epic: Financial Management

User Story 10: Configure CNO & CSR Network Fees

- **As a CNO,**
- **I want** to define the CNO and CSR session fees and percentage fees for each location,
- **so that** the financial terms are transparent and automated for every transaction.
- **Acceptance Criteria:**
 - During location setup, CNO Session Fee must default to \$0.50.
 - CNO Fee % must default to 7%.
 - CSR Session Fee and Fee % must be available for customization.
 - These values must be used in the final settlement calculations.

User Story 11: Attach Payout Account to Location

- **As a** CSO,
- **I want** to link a pre-existing payout account to a location,
- **so that** financial settlements and revenues for that site are directed to the correct account.
- **Acceptance Criteria:**
 - An "Attach Payout Account" action must be available for each location.
 - The interface should allow me to select from a list of available payout accounts from the Organization module.
 - I should be able to link multiple payout accounts to a single location.

User Story 12: Ensure Correct Fee Calculation

- **As a** CNO,
- **I want** the system to accurately calculate CNO, CSR, and net CSO fees based on the configured rates and transaction total,
- **so that** all stakeholders are compensated correctly.
- **Acceptance Criteria:**
 - The CNO Fee must be calculated as $\text{CNO Session Fee} + (\text{CNO Fee \%} * \text{Transaction Total})$.
 - The CSR Fee must be calculated as $\text{CSR Session Fee} + (\text{CSR Fee \%} * \text{Transaction Total})$.
 - The Net CSO Total must be calculated as $\text{Transaction Total} - (\text{CNO Fee} + \text{CSR Fee})$.
 - The **Transaction Total** must be the sum of Session Fee, Energy Fee, Idle Fee, and Tax.

User Story 13: Link Pricing Policy to Location

- **As a** CSO,
- **I want** any Pricing Policy I create to be tied to a specific location,
- **so that** pricing is accurately applied and financial calculations are context-aware.
- **Acceptance Criteria:**
 - When a Pricing Policy is created, it must be associated with a single Location.
 - If a policy is created from a Charge Station, the Organization and Location fields must be auto-populated and non-editable to prevent misconfiguration.

Epic: Operational Management

User Story 14: Define Location Operating Hours

- **As a CSO,**
- **I want** to define the operating hours for a location,
- **so that** end-users (drivers) are aware of the site's accessibility.
- **Acceptance Criteria:**
 - An "Operating Hours" action must be available for each location.
 - The interface must allow me to select weekdays (e.g., Monday-Friday, Select All).
 - I must be able to set a "Period Begin" and "Period End" time using time pickers with 30-minute increments.

User Story 15: Validate Operating Hours

- **As a CSO,**
- **I want** the system to prevent me from saving an operating period where the end time is before the start time,
- **so that** I can avoid creating invalid schedules.
- **Acceptance Criteria:**
 - When I set a "Period End" time that is earlier than or equal to the "Period Begin" time, the system must display a validation error.
 - The error message should clearly state: "Period End should be after Period Begin."
 - I should not be able to save the invalid time slot.

User Story 16: Manage Multiple Time Slots

- **As a CSO,**
- **I want** to define multiple, non-overlapping time slots for a single day or week,
- **so that** I can accurately represent complex operational schedules (e.g., a site closed for a lunch break).
- **Acceptance Criteria:**
 - The Operating Hours interface must allow me to "Add" another time slot for the same set of days.
 - I must be able to delete existing time slots individually.

User Story 17: Manage Site Contacts

- **As a CSO,**
- **I want** to add, view, and delete site-level contacts,
- **so that** I have an up-to-date directory of personnel responsible for each location.
- **Acceptance Criteria:**
 - A "Site Contacts" action must be available for each location.
 - The interface must allow me to add a new contact with First Name, Last Name, Contact Type, Contact Number, and Email.
 - I must be able to delete a contact from the list.

User Story 18: Auto-Populate Site Admin Contact

- **As a user,**
 - **I want** the system to automatically populate the Site Contacts list with any "Location - Site Admin" created during Charge Station setup,
 - **so that** data is consistent across modules and manual entry is reduced.
 - **Acceptance Criteria:**
 - When a charge station is set up and a "Location - Site Admin" is defined, that contact must automatically appear in the corresponding location's Site Contacts list.
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Epic: System & Non-Functional Requirements

User Story 19: Input Field Validation

- **As a user,**
- **I want** robust front-end and back-end validation on all input fields,
- **so that** I can ensure data accuracy and consistency across the system.
- **Acceptance Criteria:**
 - Mandatory fields must show an error if left blank.
 - Fields with character limits (e.g., Location Name) must show an error if the limit is exceeded.
 - Numeric fields (e.g., Utility Cost) must reject non-numeric characters.

User Story 20: Audit Trail for Location Changes

- **As a CNO administrator,**
- **I want** every action (create, update, delete) on a location to be logged,
- **so that** we have a complete audit trail for compliance and traceability.
- **Acceptance Criteria:**
 - When a location is created, a log entry must be generated with the user, timestamp, and initial data.
 - When a location is updated, a log entry must be generated showing the user, timestamp, and what fields were changed.
 - When a location is deleted, the action must be logged.

User Story 21: Secure Access to Location Data

- **As a CSO,**
- **I want** to only be able to see and manage locations that belong to my organization,
- **so that** all organizational and financial data is handled securely.
- **Acceptance Criteria:**
 - A user's access to view, create, or edit locations must be restricted based on their organizational association.

- A CSO user should not see locations belonging to another CSO in the dashboard or drop-down lists.

User Story 22: Performant Address Search

- **As a user,**
- **I want** the address search and map rendering to be fast and responsive,
- **so that** I can add and find locations without experiencing frustrating delays.
- **Acceptance Criteria:**
 - Address suggestions should appear within 500ms of me stopping typing.
 - The map should load and become interactive within 2 seconds.

User Story 23: Customize Dashboard Columns

- **As a user,**
- **I want** to show or hide columns in the Location Dashboard,
- **so that** I can create a personalized view that focuses only on the information I need.
- **Acceptance Criteria:**
 - The dashboard must have a "Show/Hide Columns" feature.
 - Selecting or deselecting a column name from the feature's menu should immediately add or remove it from the table view.
 - My column visibility preference should be saved for future sessions.

User Story 24: Refresh Dashboard Data

- **As a user,**
- **I want** a refresh icon on the dashboard,
- **so that** I can reload the latest location data without having to reload the entire page.
- **Acceptance Criteria:**
 - A refresh icon must be present on the Location Dashboard.
 - Clicking the icon should fetch and display the most recent data in the table without a full page refresh.

User Story 25: Scalable Performance

- **As a CNO,**
- **I want** the Location Module to handle a large number of organizations, locations, and stations,
- **so that** the system's performance does not degrade as our network grows.
- **Acceptance Criteria:**
 - Dashboard load time must remain under 3 seconds with 10,000+ locations in the system.
 - Filtering and searching operations must complete within 1 second regardless of the total number of locations.